BMI 500 Lecture 7: An Introduction to Model-based Machine Learning Homework

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PLEASE READ THE FOLLOWING GUIDELINES CAREFULLY BEFORE ANSWERING THE QUESTIONS.

Select only one question of your choice

- Select and deliver **only one** of the following questions (HW 1 to HW 4).
- Do not combine questions; partial answers from different questions will not be graded.

What to deliver (three items)

1. **A handwritten hypothesis page.** Every research project starts with a hypothesis that may or may not turn out to be right. Before you write any code, write your hypothesis *by hand* on one page: what you want to do and what you expect to happen in your chosen question and why. Each question lists the points your hypothesis should address. Scan or photograph the page, save it as `hypothesis.pdf` (or as an image), and make it the **first commit** of your repository, **at least 48 hours before the deadline**. Do not edit it afterwards. You are graded on the clarity of your reasoning, not on whether the hypothesis is confirmed. A rejected hypothesis or negative result with a good explanation earns full credit.[1]

2. **A codebase** in a public GitHub repository, organized by question part (e.g., `partA_...`, `partB_...`), using Python (preferably Jupyter notebooks) or MATLAB. Commit as you work and do not squash your history. The `README.md` must include:

   - your name, email, and the question number;
   - your results and answers to the discussion items, with figures (`.png` or `.jpg`);
   - a short "Discussion" (or "Hypothesis revisited") section: was your hypothesis supported, rejected, or partly supported, and why?
   - your AI statement (see below);
   - instructions to run your code (e.g., a `requirements.txt` file, or the MATLAB version and toolboxes used).

3. **A single slide** (`slide.pdf`, committed to the repository) summarizing your work: your hypothesis in one sentence and whether it was supported, your most important figure, one key number from your results, and one lesson you learned about modeling. **One slide only, not generated by AI** (even if you use AI for the problem itself; see below).

[1] In research, not every scientific hypothesis we make turns out to be correct; a well-reasoned hypothesis that the data do not support is still a valid and publishable result, often called a *negative result*. Many journals publish negative-result papers, provided the methodology is sound.

In-person presentations

After the due date, several students will be selected **at random** for a **5-minute in-person presentation** of their homework. You will use your own laptop to run your code and modify it live (e.g., change a parameter or a model component), and discuss what happens and why. Make sure your code runs on your own laptop. This is part of the deliverables, not an audit or accusation; any student may be randomly selected.

Use of AI tools

- AI tools are permitted, under two conditions:

  (a) use **a single publicly accessible AI tool** (e.g., ChatGPT, Claude, or Gemini) for the whole homework;

  (b) share a **public link to the full chat**, containing all of your interactions, as a URL in your `README.md`. Do not delete or edit any part of the conversation. If you used more than one chat in that tool, share the public links to all of them.

  Tools that leave no shareable chat record (e.g., inline code autocompletion in an editor), and chats that cannot be opened from a public URL, are not permitted. Test your link after signing out of your account to make sure that anyone can open it.

- Include one of the following statements in your `README.md`:

  – *"Statement: No generative AI (in any form) has been used to complete this homework."*

  – *"Statement: [AI-TOOL-NAME] was used to complete this homework. The full chat is available at: [PUBLIC-URL(S)]."*

- **No-AI bonus:** work completed without any generative AI receives a bonus (see the grading guidelines).

- **False claims are penalized:** a false "no generative AI" statement loses the bonus and receives a penalty (see the grading guidelines).

- **Data privacy:** all questions use synthetic or public, de-identified data. Never upload protected health information (PHI) to your repository or to an AI tool.

Note: Compliance with the above guidelines will be taken into account in grading.

HW 1: SIR and SEIR Model Implementation for Pandemic Spread

Objective: Implement and analyze the *susceptible-infectious-recovered (SIR)* compartmental model to understand the dynamics of infectious disease spread. Next, extend the model with an exposed compartment and with births and deaths. Finally, evaluate how parameter variations influence model outcomes and discuss the implications for public health strategies.
Your handwritten hypothesis should address: (i) on which day you expect the number of infected people to peak in Part B, and roughly how many people will be infected at the peak; (ii) whether you expect one or several waves of infection in Part D, and why; (iii) which parameter, β or γ, you expect to affect the peak more in Part E.

A. **Model Implementation:** Write a function in MATLAB or Python (preferably within a Jupyter notebook) that implements the SIR model:

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI,\\ \frac{dI}{dt} &= \beta SI - \gamma I,\\ \frac{dR}{dt} &= \gamma I,\end{aligned}$$

where:

- S is the number of *susceptible* people (those at risk of infection),
- I is the number of *infected* people (those currently infectious),
- R is the number of *recovered* people (those who have recovered and are immune),
- β is the *transmission rate* (per person per day), and
- γ is the *recovery rate* (per day).

Note: Implement the numerical solver yourself (e.g., forward Euler or Runge–Kutta); do not use a built-in ODE solver. You may use a built-in solver only to verify your implementation.

B. **SIR Model Simulation:** Simulate the SIR model over 160 days for a total population of $N = 2000$ people, with:

- Initial populations: $S(0) = 1995$, $I(0) = 5$, $R(0) = 0$,
- Transmission rate: $\beta = 0.125 \times 10^{-3}$ per person per day,
- Recovery rate: $\gamma = 0.1$ per day.

Plot S, I and R over time, and label each curve clearly.

C. **Analysis and Interpretation:**

i. *Infection peak:* Find the day on which $I(t)$ reaches its maximum and the number of infected people at that time. Discuss the factors that determine this peak.

ii. *Basic reproduction number:* Since S and I are counts here, the basic reproduction number is $R_0 = \beta N/\gamma$ (it reduces to β/γ when S and I are population fractions). Compute R_0 for the given parameters and interpret it. Explain how R_0 affects the spread and control of the epidemic.

iii. *Epidemic dynamics:* Describe the behavior of S, I and R over time. How many people are never infected by the end of the simulation? Discuss how the interactions between the compartments represent the spread and eventual end of the outbreak.

D. **SEIR Model with Births and Deaths:** We now add an exposed compartment and births/deaths.

i. Implement the *susceptible-exposed-infectious-recovered (SEIR)* model:

$$\begin{aligned}\frac{dS}{dt} &= \mu N - \beta SI - \mu S,\\ \frac{dE}{dt} &= \beta SI - (\sigma+\mu)E,\\ \frac{dI}{dt} &= \sigma E - (\gamma+\mu)I,\\ \frac{dR}{dt} &= \gamma I - \mu R,\end{aligned}$$

where E is the exposed (infected but not yet infectious) population, σ is the rate of becoming infectious ($1/\sigma$ is the mean latent period), and μ is the birth and death rate.

ii. Simulate for 365 days and for 1500 days with $N = 2000$, $S(0) = 1980$, $E(0) = 15$, $I(0) = 5$, $R(0) = 0$, and parameters $\beta = 0.125 \times 10^{-3}$, $\gamma = 0.1$, $\sigma = 0.25$ and $\mu = 0.005$ (all per day). Plot the compartment populations over time.

iii. Describe the pattern of infections in terms of epidemic waves. Report the timing and size of each wave.

iv. For this model, $R_0 = \dfrac{\beta N \sigma}{(\sigma+\mu)(\gamma+\mu)}$ (look up the *basic reproduction number*, or "R-naught," for its definition). Compute it and compare it with the value from Part C. Then compare the long-term number of susceptible people in your simulation with N/R_0, and explain what this number means.

v. Discuss the effect of the exposed compartment and of the birth/death rate on the epidemic dynamics. (Note: $\mu = 0.005$ per day corresponds to an average lifespan of only 200 days. This value is chosen to make the waves visible within a short simulation; discuss what would change with a realistic human lifespan.)

E. Sensitivity Analysis: Assess how parameter variations affect the outcomes and their implications for public health.

i. Run the SEIR model of Part D on a grid of parameter values: β from 0.05×10^{-3} to 0.25×10^{-3} and γ from 0.05 to 0.25 per day (at least 5 values each).

ii. For each (β, γ) pair, compute the peak number of infected people and the total number of new infections over the first 365 days. Show the results as heatmaps or contour plots, and mark the line $R_0 = 1$.

iii. Discuss the implications for public health interventions, relating β to social distancing and γ to medical treatment.

References and further reading:

1. DOI: 10.1109/JSTSP.2021.3129118
2. DOI: 10.48550/arXiv.2003.11371
3. https://github.com/alphanumericslab/EpidemicModeling

HW 2: Agent-Based Modeling of Pandemic Spread

Objective: Develop an agent-based model to simulate epidemic spread and explore the impact of interventions, such as social distancing, on infection and recovery.
Your handwritten hypothesis should address: (i) how you expect reducing the agents' movement to change the height and timing of the infection peak; (ii) whether you expect individual simulation runs to look alike, and why; (iii) whether you expect the average agent-based curves to resemble the SIR curves of the lecture.

A. Building the Base Model: Infection Dynamics in a Population

i. **Environment and initial conditions**

- Create a bounded 20×20 grid representing an area where individuals (agents) move and interact.
- Place 200 agents at random cells, in the following states:
  - 195 *susceptible* (S) agents: individuals at risk of infection;
  - 5 *infected* (I) agents: individuals who can transmit the infection;
  - 0 *recovered* (R) agents: individuals who have recovered and are immune.

ii. **Agent behaviors**

- *Movement:* At each time step, each agent moves to a neighboring cell (up, down, left or right) or stays in place, at random (a random walk). Agents cannot leave the grid. You may also try other movement rules, e.g., a Lévy walk.
- *Transmission:* If a susceptible agent shares a cell with at least one infected agent, it becomes infected with probability p.
- *Recovery:* At each time step, each infected agent recovers with probability q and becomes immune.

iii. **Simulation**

- Simulate 300 time steps, recording the number of agents in each state at each step.
- Plot the number of agents in each state over time.

iv. **Sensitivity analysis**

- Run the simulation for all combinations of $p = 0.15, 0.3$ and $q = 0.02, 0.04$.
- For each combination, run at least 20 simulations with different random initial positions. Plot the individual runs and their average for each state over time.
- Analyze how p and q affect the infection peak, the time to the peak, and the total number of agents ever infected. How much do individual runs differ from the average? In how many runs does the outbreak die out early?

B. Extending the Model: Social Distancing and Interventions

i. **Introduce social distancing measures** with $p = 0.3$ and $q = 0.02$:

- *Reduced movement:* at each step, each agent moves only with probability m (and otherwise stays in place). Which part of the model do you change to implement this?
- *Avoidance:* susceptible agents move away from cells occupied by infected agents whenever possible. Which part of the model do you change to implement this?
- For each measure, run at least 20 simulations of 300 steps and plot the average number of agents in each state over time.
- Which parameter controls the maximum number of simultaneously infected agents, which determines whether the healthcare system is overwhelmed?

ii. **Compare the results** with and without social distancing, focusing on:

- the peak number of infected agents;
- the time to the peak, and the time until no infected agents remain;
- the total number of agents ever infected;

- the implications for real-world public health interventions.

iii. **Strength of distancing:** Repeat the reduced-movement experiment with $m = 1.0, 0.6, 0.3$ and 0.1. Plot the average peak number of infected agents and the time to the peak against m, and discuss how stricter or more relaxed distancing affects the epidemic.

References and further reading:

1. https://youtu.be/gxAaO2rsdIs
2. DOI: 10.1109/JSTSP.2022.3145622
3. https://youtu.be/pasyQympFGE
4. https://github.com/alphanumericslab/EpidemicModeling

HW 3: Model-based Bias Removal in Machine Learning using Synthetic Blood Pressure Data

Objective: Model age-related trends in blood pressure; then explore how imbalanced datasets bias a classifier and evaluate mitigation methods. You will generate synthetic systolic (SBP) and diastolic (DBP) blood pressure data for males and females from a bivariate normal model, train a classifier to predict sex from SBP and DBP at different male/female ratios, evaluate its performance, and propose and test bias mitigation strategies.
Your handwritten hypothesis should address: (i) which model form (polynomial or sigmoid/Gaussian) you expect to fit SBP and DBP better, and why; (ii) how you expect the overall accuracy and the per-sex recall to change as the training set becomes more imbalanced; (iii) what you expect a classifier trained on a 90% male data set to do for female patients.

A. Modeling Blood Pressure as a Function of Age: Figures 3 and 4 of the paper by S. Mousavi et al. [paper] show age-related trends in SBP and DBP: SBP typically increases with age and may level off at older ages, while DBP peaks around middle age and then stabilizes or declines.

i. **Mathematical models:** Fit the two model families below (or propose your own), where a is age in years.

- *Polynomial model:*

$$\mathrm{SBP}(a) = c_1 a^2 + c_2 a + c_3, \qquad \mathrm{DBP}(a) = d_1 a^2 + d_2 a + d_3.$$

- *Sigmoid model for SBP and Gaussian model for DBP* (each with a baseline, since blood pressure never approaches zero):

$$\mathrm{SBP}(a) = S_0 + \frac{\Delta S}{1 + e^{-k(a-a_0)}}, \qquad \mathrm{DBP}(a) = D_0 + \Delta D\, e^{-\frac{(a-a_{\mathrm{peak}})^2}{2\sigma^2}}.$$

ii. **Model fitting:** Use the numeric tables of mean SBP and DBP by age in the [paper]. Fit c_1, c_2, c_3 and d_1, d_2, d_3 by linear least squares, and $S_0, \Delta S, k, a_0$ and $D_0, \Delta D, a_{\mathrm{peak}}, \sigma$ by nonlinear least squares, in Python or MATLAB. Fit males and females separately.

iii. **Quantitative evaluation:** Compute the mean squared error (MSE) and R^2 of each model. Since the models have different numbers of parameters, also compare them with the Akaike information criterion (AIC), the minimum description length (MDL), or the Bayesian information criterion (BIC). Plot the fitted curves over the data points.

iv. **Interpret the model parameters:**

- Polynomial model: interpret c_1 and d_1 (the curvature terms) and give the physical dimension (units) of every coefficient.
- Sigmoid/Gaussian models: interpret S_0 (baseline SBP at young ages), ΔS (total rise of SBP with age), a_0 (age at half of the rise), k (steepness of the rise), D_0 (baseline DBP), ΔD (height of the DBP peak above baseline), a_{peak} (age of peak DBP) and σ (width of the DBP peak), with their units.

v. **Discussion:**

(a) Which model captures the age trends of SBP and DBP better? Does the answer change when the number of parameters is taken into account (AIC, MDL or BIC)?

(b) How do the fitted parameters reflect physiological changes of blood pressure with age, and how do they differ between males and females?

(c) How do the models behave outside the age range of the data? Discuss their limitations in capturing demographic differences.

B. Model-based Bias Removal using Synthetic Blood Pressure Data: We now study how dataset imbalance affects a classifier that predicts sex from SBP and DBP, and how synthetic data can help mitigate the resulting bias.

i. **Synthetic data generation**

- Using a bivariate normal model, generate synthetic SBP and DBP data with the following statistics:
  - **Male:** SBP mean $\mu_{sbp}^{(M)} = 133.0\,\text{mmHg}$, SD $\sigma_{sbp}^{(M)} = 18.6\,\text{mmHg}$; DBP mean $\mu_{dbp}^{(M)} = 78.8\,\text{mmHg}$, SD $\sigma_{dbp}^{(M)} = 12.6\,\text{mmHg}$; correlation $\rho^{(M)} = 0.45$.
  - **Female:** SBP mean $\mu_{sbp}^{(F)} = 125.8\,\text{mmHg}$, SD $\sigma_{sbp}^{(F)} = 19.0\,\text{mmHg}$; DBP mean $\mu_{dbp}^{(F)} = 74.8\,\text{mmHg}$, SD $\sigma_{dbp}^{(F)} = 12.4\,\text{mmHg}$; correlation $\rho^{(F)} = 0.50$.
- Generate data sets of 60,000 samples with M male and F female entries ($M + F = 60{,}000$), for the male/female ratios 50/50, 75/25, 90/10 and 97/3.
- Label males as 1 and females as 0.
- Plot the two distributions (e.g., scatter plots with the 95% probability ellipses) and comment on their overlap.

ii. **Binary classification**

- **Data splitting:** Split each data set into training (70%) and test (30%) sets, stratified by sex. In addition, create one *balanced* test set (50/50) of 10,000 new samples.
- **Model training:** Train a binary classifier (e.g., logistic regression, SVM, or a decision tree) that predicts sex from SBP and DBP only, on each training set.
- **Evaluation:** For each classifier, on both its own test set and the balanced test set:
  (a) plot the ROC curve and report the area under it (AUC);
  (b) report the accuracy, the F1 score, and the recall of each sex (the fraction of males, and of females, classified correctly).
- **Analysis:**
  (a) Discuss how performance changes with the male/female ratio. Why can accuracy increase while the classifier becomes less useful for one group?
  (b) Why does the AUC change much less than accuracy and recall?
  (c) Identify the biases introduced by the different prevalences.

iii. **Discussion**

(a) Reflect on the importance of balanced datasets in healthcare.
(b) Discuss the implications and challenges of real-world imbalanced datasets.
(c) Suggest strategies to address these challenges in practice, based on the examples given in class.

iv. **Bias mitigation in training:** Using the 90/10 training set, modify the training to reduce sex bias. At least one method should be the bias-penalized loss of the lecture, $C(\theta) = f(\theta) + \lambda\,\phi(\theta)$, where f is the usual loss and ϕ is the absolute difference between the recalls of the two sexes. (Since ϕ is not differentiable, you may use a smooth approximation, or a grid search over class weights and the decision threshold; explain your choice.) Sweep λ and plot the overall accuracy against ϕ on the balanced test set (the accuracy–equity trade-off curve). Explain the reasoning behind your method.

v. **Synthetic augmentation:** Augment the 90/10 training set with synthetic female samples drawn (a) from the female distribution above and (b) uniformly over a wide SBP/DBP range covering the corner cases. Compare the performance and bias of the two choices on the balanced test set. Which is better for performance? Which do you consider more ethical, and why?

References and further reading:

1. https://arxiv.org/pdf/2306.08451.pdf
2. https://arxiv.org/pdf/2402.01598
3. https://doi.org/10.1016/j.jelectrocard.2022.07.007

HW 4: Biopotential Modeling and Synthetic ECG Generation

Objective: Understand and simulate biopotential signals, from action potential models to ECG generation. We explore dynamic models of oscillations, apply the McSharry–Clifford synthetic ECG model, and evaluate multichannel signal generation, with a view to their use in machine learning.
Your handwritten hypothesis should address: (i) how you expect the solution of the second-order oscillator to change as α goes from positive to zero to negative; (ii) whether you expect the amplitude of the Van der Pol oscillator to depend on its initial condition, and how its period will change as α grows; (iii) which McSharry–Clifford parameters you expect to change the QRS complex and which the T wave.

A. Action Potential Dynamics and Oscillations:

i. **Second-order linear oscillator:** The differential equation

$$\ddot{x}(t) + 2\alpha\dot{x}(t) + \omega_0^2 x(t) = 0$$

is the simplest model of an oscillation. Use $\omega_0 = 2\pi/0.8 \approx 7.85\,\mathrm{rad/s}$ (a period of 0.8 s, i.e., 75 beats per minute) and $x(0) = 1$, $\dot{x}(0) = 0$. Implement the equation in Python or MATLAB and simulate the following cases:

- $0 < \alpha < \omega_0$ (e.g., $\alpha = 0.5$ and $2\,\mathrm{s}^{-1}$): underdamped, decaying oscillations;
- $\alpha = \omega_0$: critically damped, no oscillation;
- $\alpha > \omega_0$ (e.g., $\alpha = 15\,\mathrm{s}^{-1}$): overdamped, no oscillation;
- $\alpha = 0$: undamped, sustained oscillation;
- $\alpha < 0$ (e.g., $\alpha = -0.3\,\mathrm{s}^{-1}$): unstable, growing oscillations.

Plot the time-domain signals and the phase-plane trajectories ($\dot{x}(t)$ vs. $x(t)$) for each case and interpret the differences.

ii. **Van der Pol oscillator (self-regulating dynamics):** Implement the Van der Pol model, a nonlinear extension of the oscillator above:

$$\ddot{x}(t) - 2\alpha[1 - x(t)^2]\dot{x}(t) + \omega_0^2 x(t) = 0,$$

with $\omega_0 = 7.85\,\mathrm{rad/s}$ and $\alpha = 0.5$, 5 and $20\,\mathrm{s}^{-1}$.

- Start from one initial condition inside the limit cycle (e.g., $x(0) = 0.1$) and one outside it (e.g., $x(0) = 4$), with $\dot{x}(0) = 0$. Plot the phase-plane trajectories and show that both converge to the same limit cycle.
- Measure the amplitude and the period of the steady oscillation for each α. How do they compare with the linear oscillator's period $2\pi/\omega_0 = 0.8\,\mathrm{s}$?
- Explain how the feedback mechanism (the damping coefficient becoming a function of $x(t)$) stabilizes the amplitude, and how this model differs from the linear oscillator.

iii. **Comparing the FitzHugh–Nagumo and Van der Pol models:** Implement the FitzHugh–Nagumo model

$$\begin{cases} \dfrac{dv}{dt} = v - \dfrac{v^3}{3} - w + I, \\ \dfrac{dw}{dt} = \epsilon\,(v + a - bw), \end{cases}$$

with the classic parameters $a = 0.7$, $b = 0.8$ and $\epsilon = 0.08$ (dimensionless time).

- Simulate the model for $I = 0$ and $I = 0.5$, and plot the time-domain waveforms and phase-plane trajectories together with the two nullclines ($\dot{v} = 0$ and $\dot{w} = 0$).
- Sweep I from 0 to 1.6 and find the approximate range of I for which sustained oscillations (repetitive firing) occur.
- Compare the waveforms and phase portraits with those of the Van der Pol oscillator. Discuss the advantages and limitations of each model for capturing the dynamics of action potentials.

B. Synthetic ECG Generation Using the McSharry–Clifford Model:

i. **Generating synthetic ECG signals:** Using the OSET toolbox (available here), or your own implementation in Python or MATLAB, implement the original McSharry–Clifford synthetic ECG model (or one of its extensions in OSET):

$$\begin{aligned}\dot{x} &= \rho x - \omega y,\\ \dot{y} &= \rho y + \omega x,\\ \dot{z} &= -\sum_i a_i \Delta\theta_i \exp\left(-\frac{\Delta\theta_i^2}{2b_i^2}\right) - (z - z_0)\end{aligned}$$

Use the parameters of the original paper (DOI: 10.1109/TBME.2003.808805) or of the polar extension in OSET, at a mean heart rate of 70 beats per minute. Plot 10 s of synthetic ECG in the time domain and analyze its morphology in the phase plane.

ii. **Parameter variation:** Change the amplitudes a_i and widths b_i of the P, QRS and T kernels one at a time, and show how each change affects the ECG morphology. Discuss the observed effects.

iii. **Heart rate variability (HRV):** Make ω a function of time to simulate a changing heart rate: use a mean heart rate of 70 beats per minute with a sinusoidal modulation of ± 4 beats per minute at 0.25 Hz (the breathing rate of 15 breaths per minute, mimicking respiratory sinus arrhythmia). Plot the modified ECG and the resulting RR-interval series, and discuss the effects of HRV on the signal.

C. Multichannel ECG Modeling and Stochastic Extensions: Run the multichannel ECG simulation codes in OSET, including stochastic variability and maternal–fetal ECG modeling, and report on their performance. Add random noise and small random deviations of the kernel parameters to mimic beat-to-beat variability. Discuss why stochastic modeling is important for generating realistic ECG signals for machine learning.

References and further reading:

1. McSharry, P. E., Clifford, G. D., Tarassenko, L., & Smith, L. A. (2003). A dynamical model for generating synthetic electrocardiogram signals. *IEEE Transactions on Biomedical Engineering*, 50(3), 289–294. DOI: 10.1109/TBME.2003.808805

2. OSET toolbox: https://github.com/alphanumericslab/OSET

3. FitzHugh, R. (1961). Impulses and physiological states in theoretical models of nerve membrane. *Biophysical Journal*, 1(6), 445–466.

4. Sameni, R., Clifford, G. D., Jutten, C., & Shamsollahi, M. B. (2007). Multichannel ECG and noise modeling: application to maternal and fetal ECG signals. *EURASIP J. Adv. Signal Process.* DOI: 10.1155/2007/43407

Grading Guidelines for Homework Evaluation

Each submission is evaluated on the following factors:

1. **Methodology (30%)**: Implementation correctness, adherence to the question's requirements, and suitability of the methods:
   - Implementation follows the mathematical models and equations as described.
   - Appropriate numerical techniques are chosen (e.g., self-implemented ODE integration, random walks, least squares, Python or MATLAB packages).
   - Logical flow of the solution and validation of results with intermediate checks.
2. **Completeness (25%)**: All parts of the selected question are addressed:
   - All sub-parts are completed with the required simulations, plots and analyses.
   - Edge cases and sensitivity analyses are explored as required.
   - All major parts (A, B, C, ...) carry equal weight.
3. **Hypothesis (10%)**: The handwritten hypothesis page and the "Discussion" (or "Hypothesis revisited") section:
   - The page is handwritten, addresses the listed points, and is the first commit of the repository, made at least two days before the deadline.
   - The hypothesis is specific and reasoned.
   - The discussion honestly compares the hypothesis with the results and explains any differences. Grade the reasoning, *not* whether the hypothesis was confirmed.
4. **Documentation and Single-Slide (15%)**:
   - Code is clear and commented (short comments are acceptable), and runs with the provided instructions.
   - The README includes results, answers and well-labeled figures.
   - The single slide contains the hypothesis, its verdict, a key figure, a key number and a lesson learned, and is clear, readable and not generated by AI.
5. **Critical Analysis (10%)**: Discussion and interpretation:
   - Key results are explained with context and implications.
   - Comparisons are drawn between models or scenarios.
   - Limitations and suggestions for improvement are included.
6. **Adherence to Guidelines (10%)**:
   - All three deliverables are present (hypothesis page, codebase, single slide).
   - Commit history is preserved, with the hypothesis page as the first commit, at least two days before the deadline.
   - AI statement included; if AI was used, a single tool was used and the public chat link(s) are shared.
7. **Novelty (Bonus, up to 5%)**: Creative or innovative work beyond the question, such as additional scenarios, insightful visualizations, or techniques not explicitly required.
8. **No-AI Bonus (5%)**: For homework completed without any generative AI tool, with the corresponding statement in the README.
9. **Penalty for a false no-AI statement (-25%)**: A "no generative AI" statement that is found to be false (e.g., during grading or the in-person presentation) loses the no-AI bonus, receives a 25% penalty on the total grade, and may be handled under the Emory Honor Code.

In-person presentations: Randomly selected students present for 5 minutes on their own laptops: they run their code and modify it live, and discuss it. The presentation confirms the student's contribution and understanding. Parts of the homework that the student cannot run, modify or explain may lose credit.

Notes for course TA:

- If you suspect a false "no generative AI" statement, or the use of more than one AI tool, do not apply the penalty yourself. Please note it in the "Notes" column of the grading spreadsheet so that I can check it.
- Check that the hypothesis page is handwritten and is the first commit, and that the public AI chat links (if any) open and contain the full conversation. Also, compare the dates and times of the chat logs with the GitHub commit history.
- Please write short, fact-based feedback on each part that I can review and share with the students. I will personally review the deliverables after your grading. Thank you.

Grading Rubric

Table 1: Grading rubric for homework evaluation

Criteria	**Poor (0–4)**	**Fair (5–7)**	**Excellent (8–10)**
Methodology	Incorrect equations or methods; incomplete code.	Partial implementation; minor issues with logic or results.	Fully correct and well-validated implementation.
Completeness	Significant missing parts; edge cases not explored.	Most parts completed; some sensitivity analyses or scenarios missing.	All parts addressed, including edge cases and full analysis.
Hypothesis	Missing, typed, not the first commit, committed late, or vague.	Handwritten and on time, but generic or weakly reasoned; shallow revisit.	Specific, well-reasoned hypothesis; honest and insightful revisit.
Documentation and Slide	Uncommented code that does not run; no clear README; missing or cluttered slide.	Basic README and comments; slide missing some required elements.	Clear README and code that runs; a clear single slide with all required elements.
Critical Analysis	Lacking or superficial explanations.	Basic explanations with limited depth.	Thoughtful, in-depth discussion of results and limitations; aspects beyond the question.
Adherence to Guidelines	Major deviations (e.g., missing deliverables, no AI statement, multiple AI tools, missing chat links).	Minor deviations (e.g., incomplete run instructions, broken chat link).	All instructions followed, including commit order and AI policy.
Novelty (Bonus)	None or trivial additions.	Basic extensions.	Significant creative additions or advanced analyses.